
LilNetX: Lightweight Networks with EXtreme Model Compression and Structured Sparsification

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Abstract

We introduce LilNetX, an end-to-end trainable technique for neural networks that enables learning models with specified accuracy-rate-computation trade-off. Prior works approach these problems one at a time and often require post-processing or multistage training which become less practical and do not scale very well for large datasets or architectures. Our method constructs a joint training objective that penalizes the self information of network parameters in a reparameterized latent space to encourage small model size while also introducing priors to increase structured sparsity in the parameter space to reduce computation. We achieve up to 50% smaller model size and 98% model sparsity on ResNet-20 while retaining the same accuracy on the CIFAR-10 dataset as well as 35% smaller model size and 42% structured sparsity on ResNet-50 trained on ImageNet, when compared to existing state-of-the-art model compression methods. Code is available at <https://github.com/Sharath-girish/LilNetX>.

1 Introduction

Recent research in deep neural networks (DNNs) has shown that large performance gains can be achieved on a variety of computer vision tasks simply by employing larger parameter-heavy and computationally intensive architectures [13, 26]. However, as the DNNs proliferate in the industry, they often need to be trained repeatedly, transmitted over the network to different devices, and need to perform under hardware constraints with minimal loss in accuracy, all at the same time. Hence, finding ways to reduce the storage size of the models on the devices while simultaneously improving their run-time is of utmost importance. This paper proposes a general purpose neural network training framework to jointly optimize the model parameters for accuracy, model size on the disk and computation, on any given task.

Over the last few years, the research on training smaller and efficient DNNs has followed two seemingly parallel tracks with different goals: One line of work focuses on model compression to deal with the storage and communication network bottlenecks when deploying a large number of models over the air. While they achieve high levels of compression in terms of memory, their focus is not on reducing computation. They either require additional algorithms with some form of post hoc training [71] or quantize the network parameters at the cost of network performance [10, 39]. The other line of work focuses on reducing computation through various model pruning techniques [16]. The focus of these works is to decrease the number of Floating Point Operations (FLOPs) of the network at the inference time, albeit they are also able to achieve some compression due to fewer parameters. Typically, the cost of storing these pruned networks on disk is much higher than the dedicated model compression works.

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